Reg.No :.....

Name:.....

MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)

THIRDSEMESTEREXAMINATION

(2024ADMISIONONWARDS)

MG3DSCECT202 -Internet ofThings

Duration:1.5hours **MaximumMarks:50**

**Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S),Interest (I)andAppreciation(Ap)*

Part A

MultipleChoiceQuestions
Answer All Questions
Eachquestioncarries1marks
(Writetheoptionsandanswerwhileansweringeachquestion)

1. Whichofthefollowing isawidely usedembedded platform for IoTprototyping? K 1
   A. MicrosoftWord B. Adobe Photoshop
   C.ArduinoUNO D.CorelDRAW
2. WhyisESP32preferredinIoTapplicationsovertraditionalmicrocontrollers? U 1
   A. Ithasabuilt-inprinter B.Itconsumesmorepower
   C. ItsupportsWi-Fiand Bluetoothconnectivity
   D. Itdoesnotsupportanyprogramming
3. WhichsoftwareiscommonlyusedtoprogramtheESP32 microcontroller? K 1
   A.Turbo C++ B.ArduinoIDE C. Notepad D.MSExcel
4. YouwanttoblinkanLEDconnectedtoGPIOpin5oftheESP32.WhichlineofArduino code should you write inside the setup() function? A 2
   A. digitalWrite(5, HIGH); B. digitalRead(5);
   C.pinMode(5, OUTPUT); D.analogWrite(5,255);
5. AnLDRisconnectedtoananalogpinontheESP32.Toreaditsvalue,whichArduino function should you use? A 2
   A. analogRead(pin); B.digitalWrite(pin,HIGH);
   C.analogWrite(pin, value); D.pinMode(pin,INPUT);
6. Whyisthe Serial.begin(9600);statementused inanESP32 sketch? U 2
   A. Toturn onthe Wi-FimoduleB. Toset upthespeedofanalog pins
   C. Toinitializeserialcommunicationatabaudrateof9600
   D. TostartGPIO operation
7. Whichofthe followingisanexampleofanoutput device

commonlyusedwithESP32? K 2

A.PIRSensor B.Rain Sensor C.UltrasonicSensorD. Buzzer

8. Youaredevelopinga project wheretheESP32must send sensordatatoalocalwebserver.WhichmodeshouldtheESP32operateintoconnecttoan existing Wi-Fi network? A 3

A. AccessPointMode B.SleepMode

C. Station Mode D.OfflineMode

9. WhatdoesBLEstandforinthecontextofESP32? K 3

A. BasicLAN Extension B.BluetoothLowEnergy

C.BoardLevelExecution D.BinaryLink Encoder

10. WhatisthemaindifferencebetweenanHTTPserverandanHTTP clientinanESP32webapplication? U 3

A. HTTPserverstoresdatapermanentlywhile clientdeletesit

B. HTTPclientrequests data;HTTPserverrespondswithdata

C. HTTPservergeneratespower;clientusesit

D. HTTPclientandserverarebothusedonlyinBLE mode

10*1 = 10

PartB

ShortAnswerQuestions.Answer**4**Questions.
EachQuestioncarries5marks

11. ExplainthebasiccomponentsofanIoTsystemwithaneatblockdiagram. Includekeydevices andtheirrolesinthearchitecture. U 1

12. ListandbrieflydescribeanyfourmajorchallengesfacedinimplementingIoTin modern society. K 1

13. Write an Arduino code to blink an LED connected to a digital GPIO pin of an ESP32 board. Also explain the pin configuration used. A 2

14. Differentiate between analogRead() and analogWrite() functions in Arduino programmingwithexamples. U 2

15. Defineany3sensors usedinIoTapplications andstateoneapplicationofeach.K 2

16. Writeaprogram to display temperaturesensor data (DHT11) on an LCDusing ESP32. Brieflyexplainhowthesensordataisreadanddisplayed. A 3

5*4 = 20

PartC

EssayQuestions.Answer**any2**Questions Each question carries 10 marks

17. DifferentiatebetweenArduinoandESP32platformsintermsoffeaturesandusageinIoT applications. U 1

18. Designasmallproject toturnonanLEDwhenthePIRsensordetects motion. A 2

19. Buildabasiccircuitwherepressingabuttonactivatesa buzzerusing Arduino. A 2

20. DifferentiatebetweenStationmodeandAccessPointmodeinESP32Wi-Finetworking. U 3

10*2 = 20